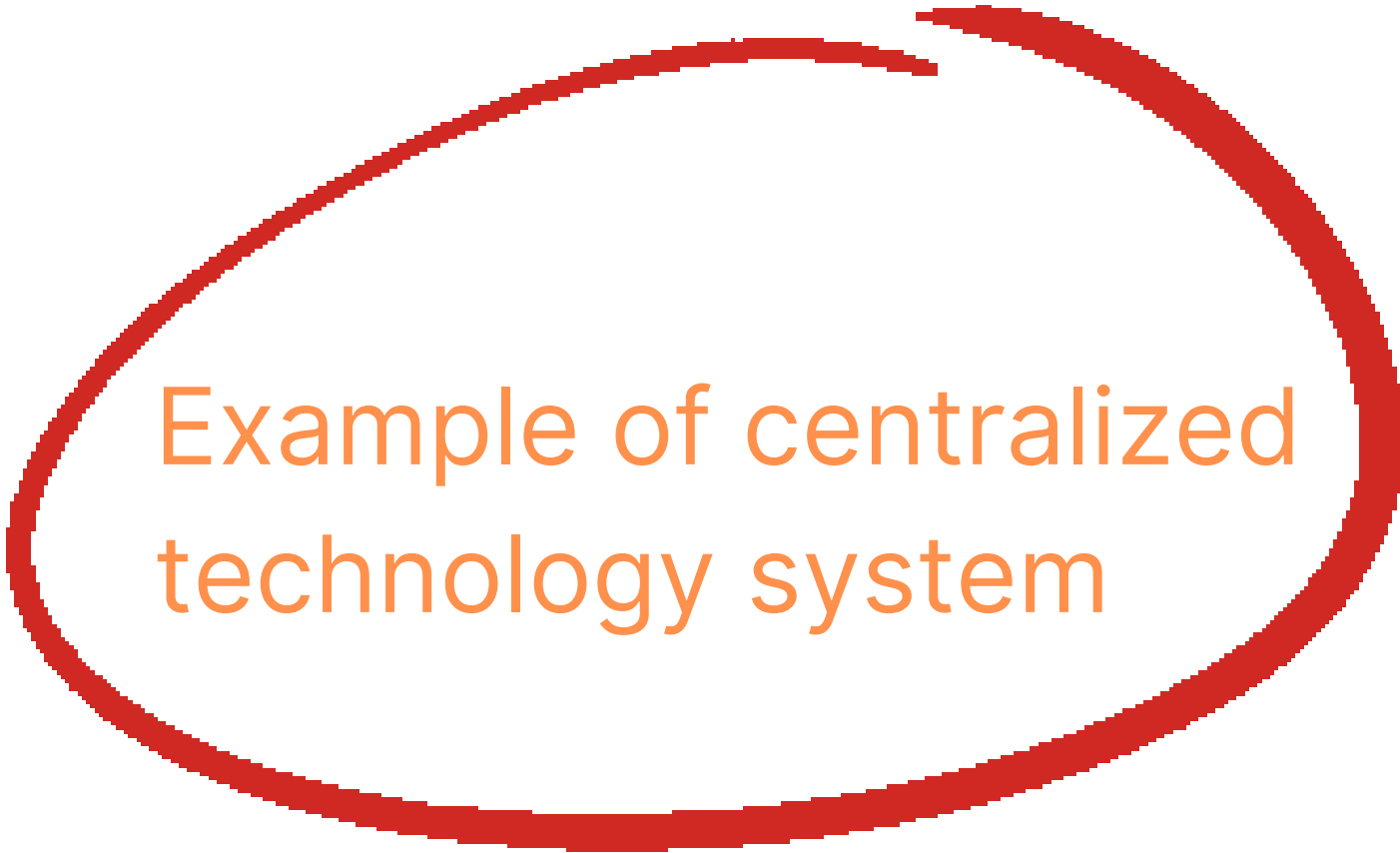




BLOCKCHAIN TECHNOLOGY

PROBLEM OF CENTRALIZED TECHNOLOGY

- 01 Security
- 02 Single point of failure
- 03 Data privacy
- 04 Transparency



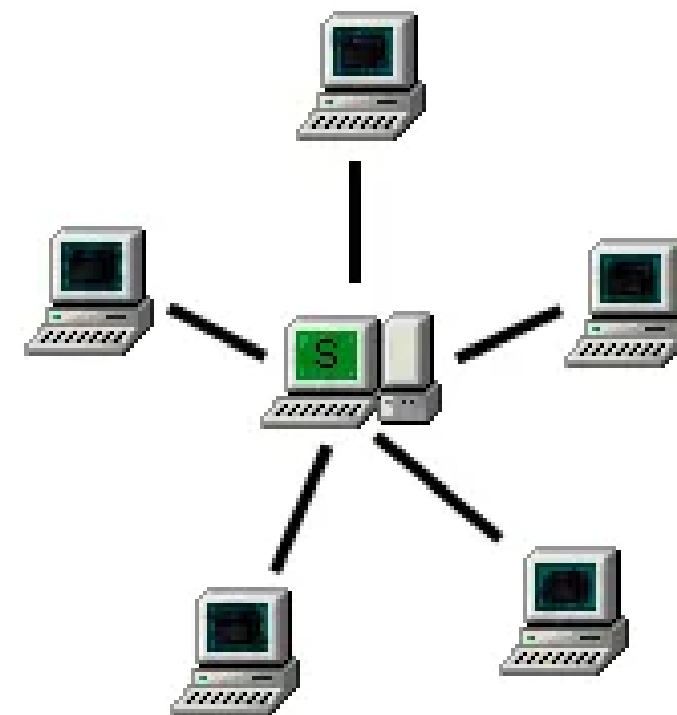
Example of centralized
technology system

Technologies behind blockchain

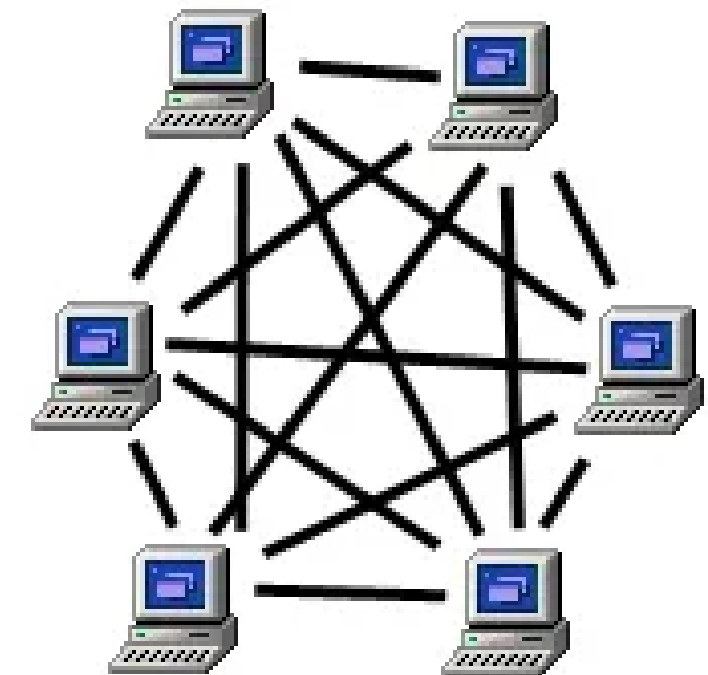
- 01 Peer-to-peer network
- 02 Distributed ledger technology
- 03 Cryptography

Example of peer-to-peer
market

Server Based Network



Peer to Peer Network



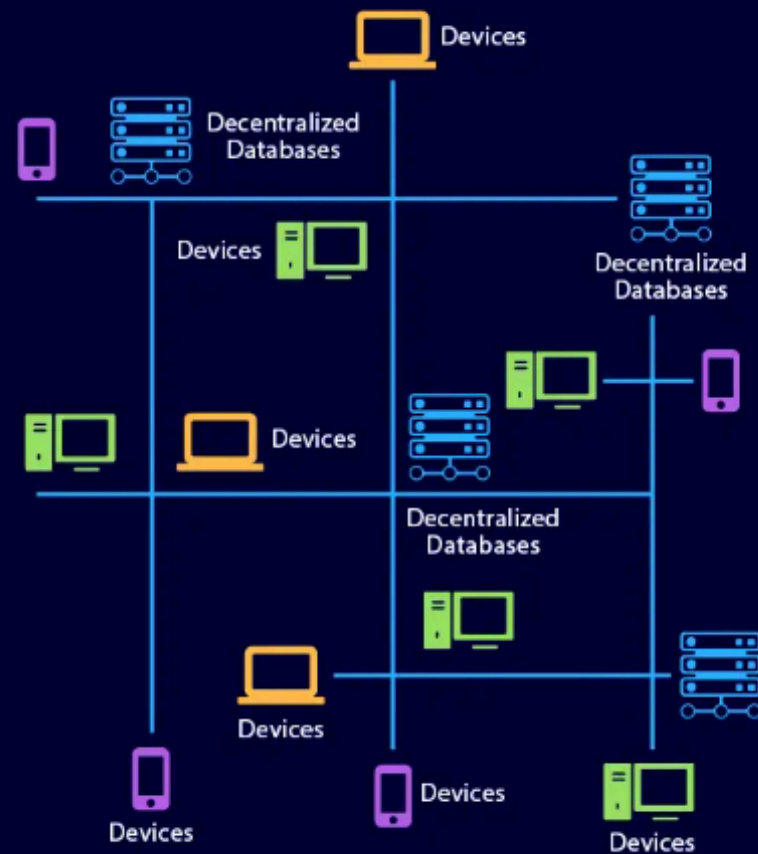
Centralized -- Decentralized -- Distributed

Centralized vs Decentralized Internet

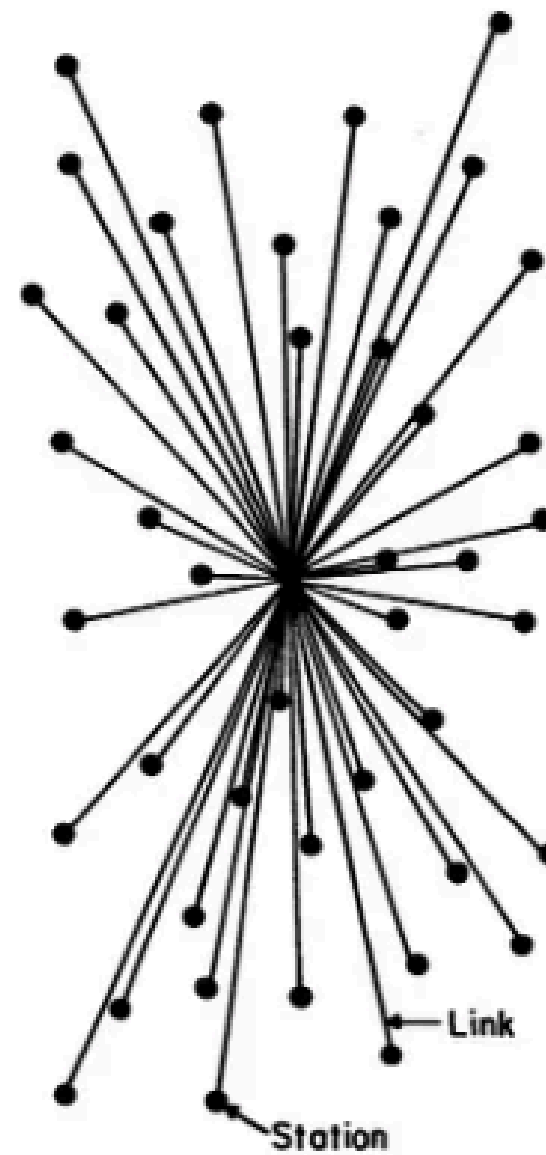
BEFORE



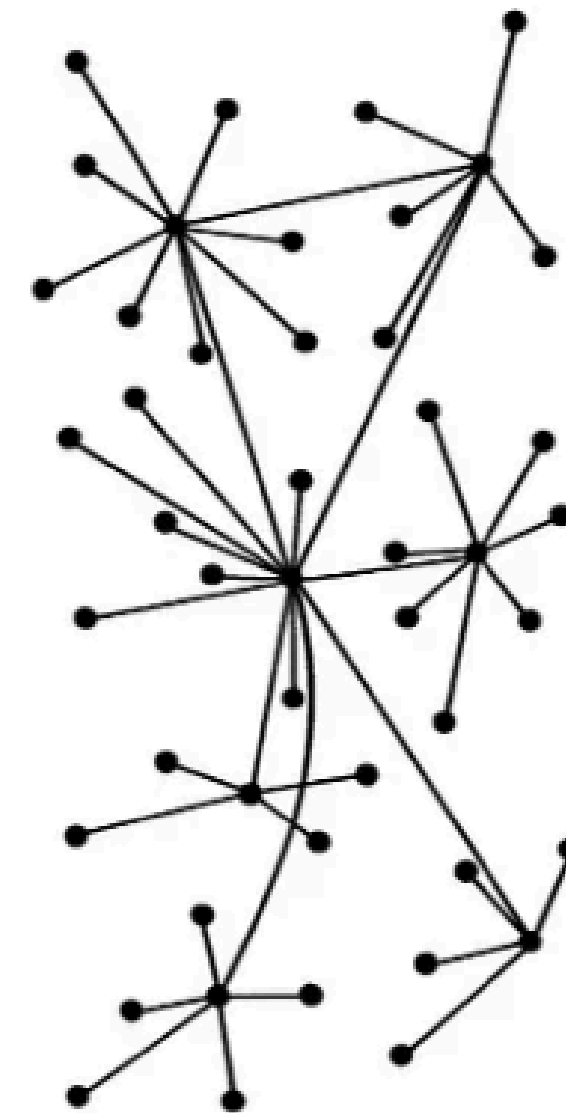
AFTER



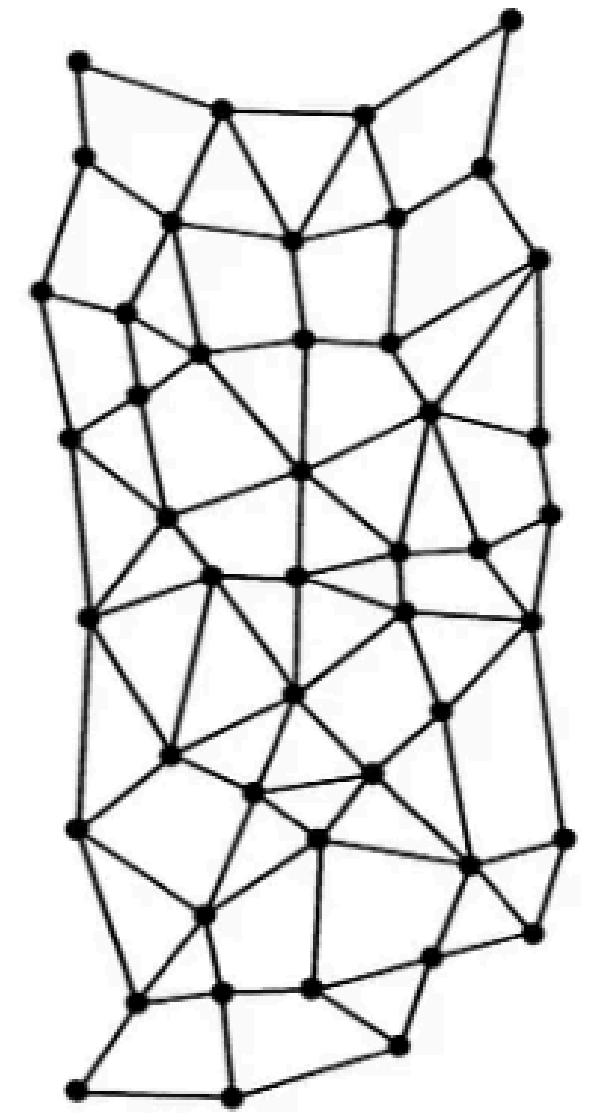
 **101 Blockchains**
Created by 101blockchains.com



CENTRALIZED
(A)



DECENTRALIZED
(B)



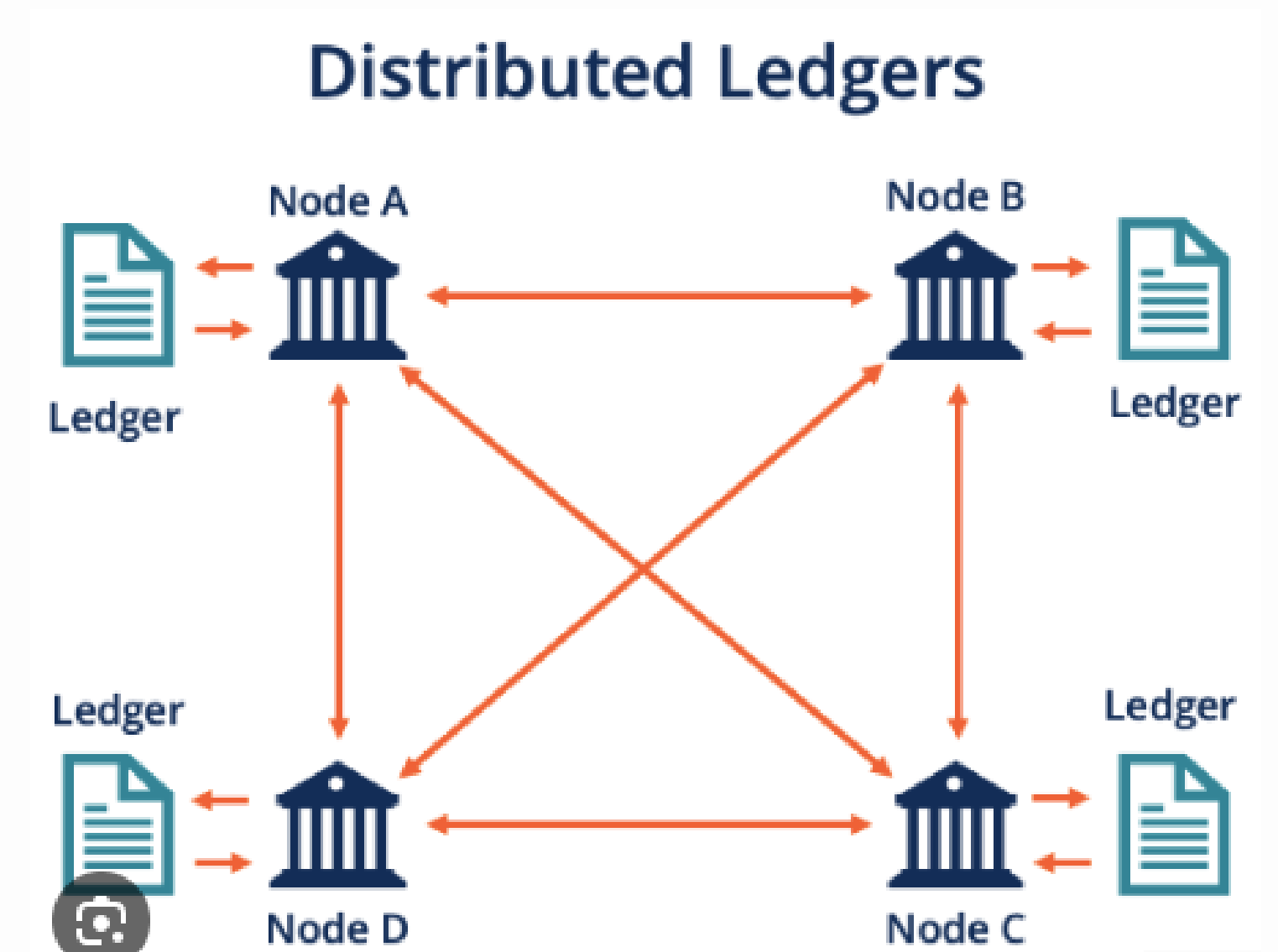
DISTRIBUTED
(C)

Distributed Ledger Technology

Distributed Ledger Technology (DLT) เป็นการบันทึกข้อมูลและธุรกรรมแบบออนไลน์โดยข้อมูลจะถูกจัดเก็บไว้หลาย ๆ ที่ในเวลาเดียวกัน

คุณลักษณะของ DLT

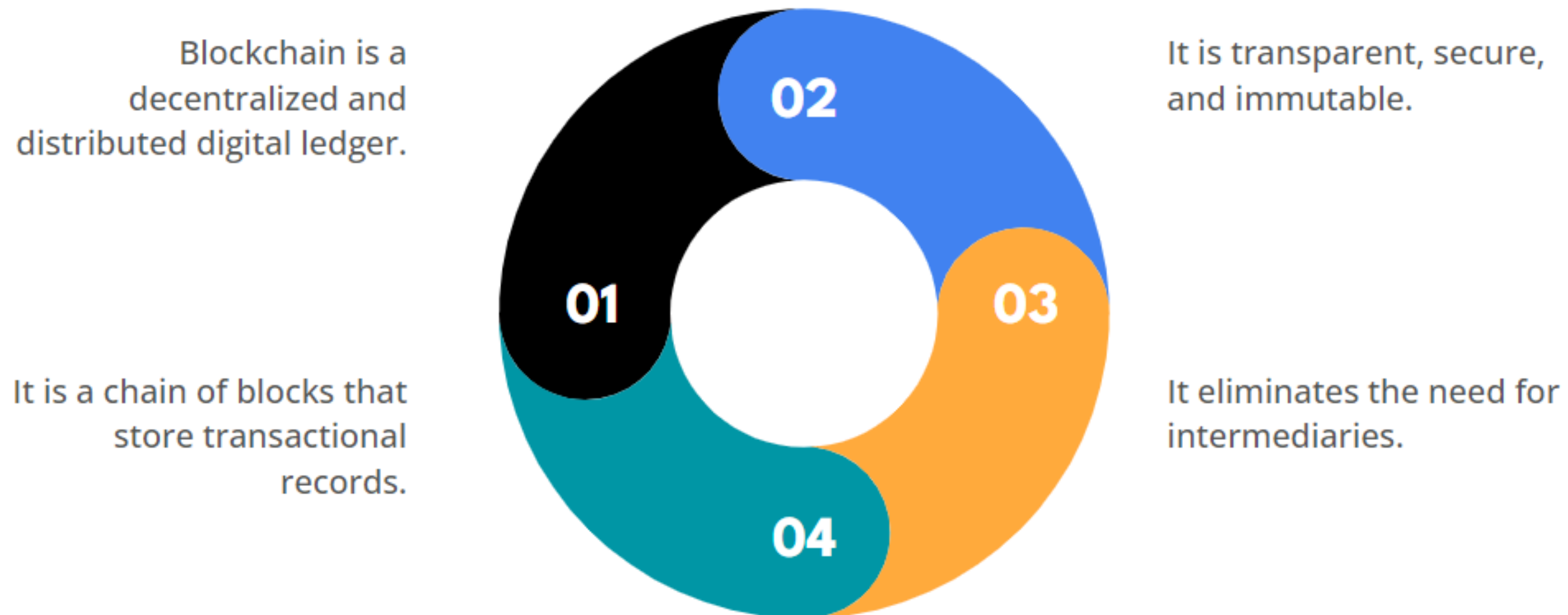
- 01 การกระจายข้อมูล (decentralization)
- 02 ฉันทามติ (consensus)
- 03 ความโปร่งใส (transparency)
- 04 ความปลอดภัย (security)



BLOCKCHAIN

Blockchain (บล็อกเชน) คือ ระบบการเก็บข้อมูลแบบกระจาย (แบบหนึ่งของ DLT) ที่มีความน่าเชื่อถือ โปร่งใส และไม่ต้องอาศัยคนกลาง

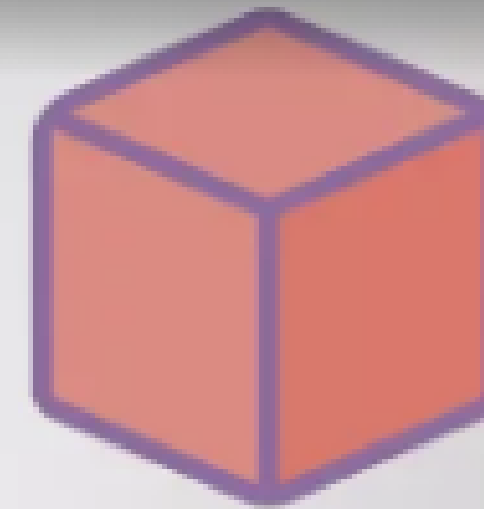
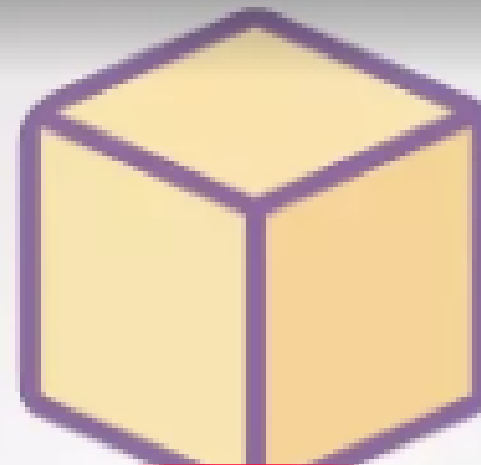
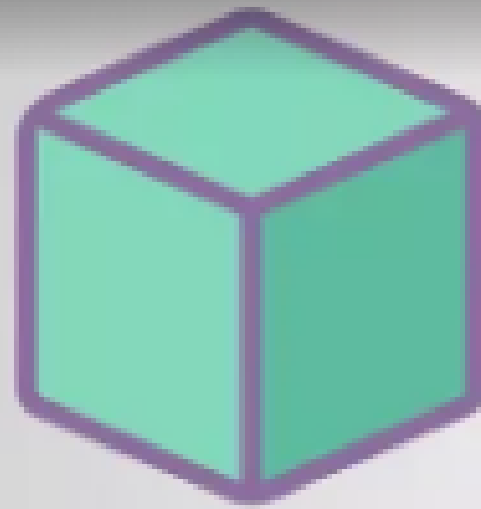
What is Blockchain?





How does a blockchain work - Simply Explained

Simply Explained

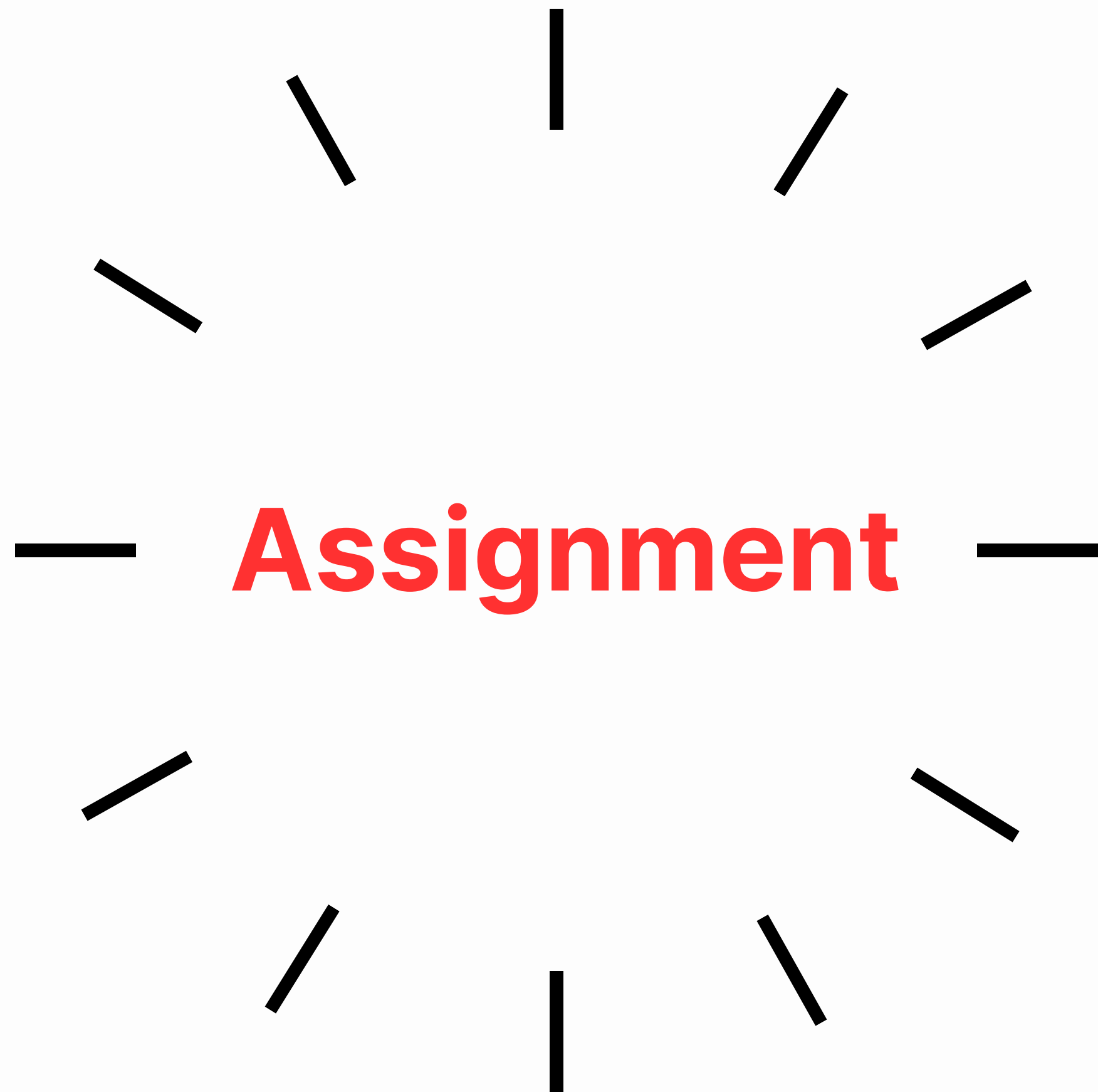


Blockchain

— *Simply explained* —



Watch on  YouTube



Cryptography

Cryptography คือ เทคโนโลยีที่เกี่ยวข้องกับการป้องกันข้อมูลและความเป็นส่วนตัวโดยใช้เทคนิคการเข้ารหัสข้อมูล

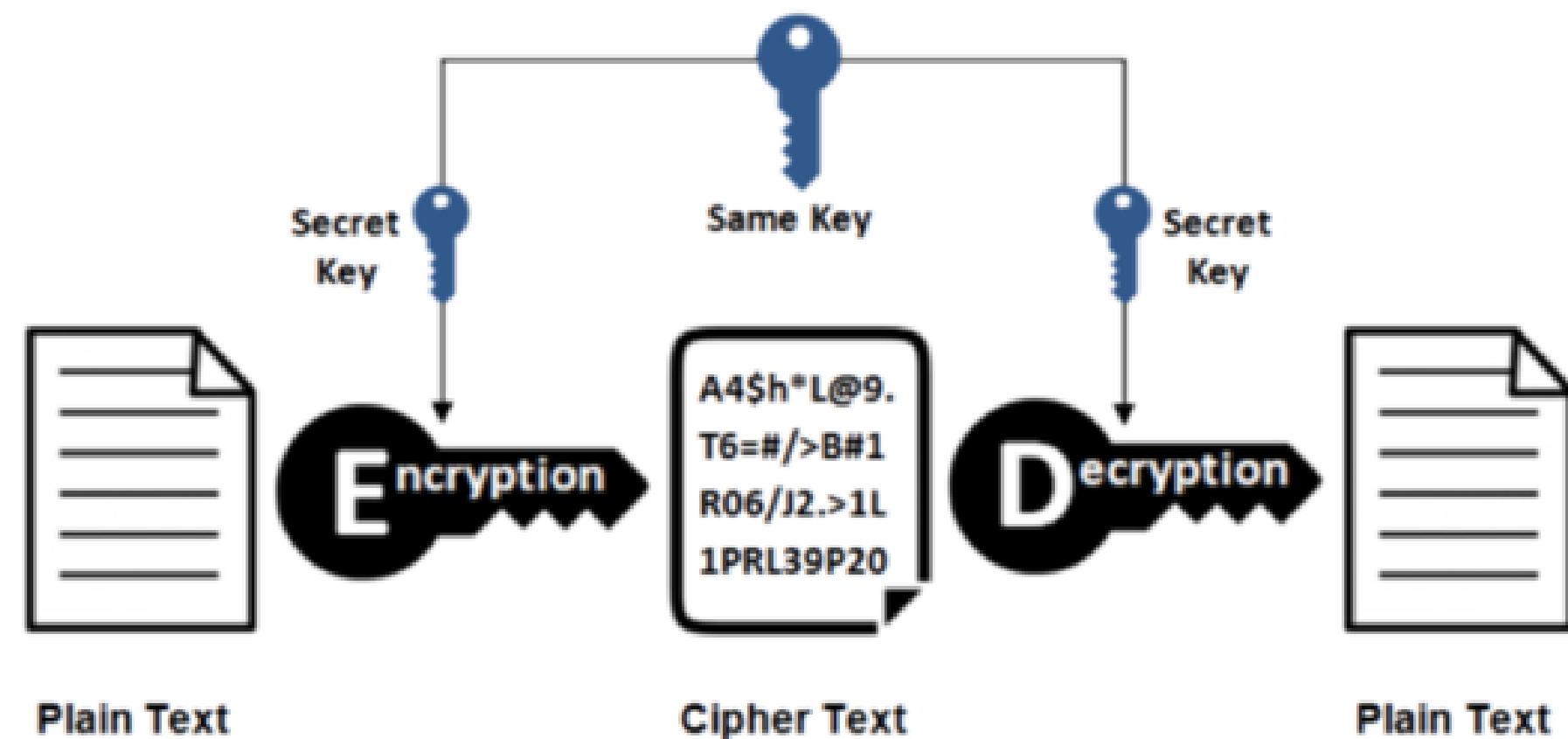
Cryptographic techniques used in blockchain

1) Public-key cryptography

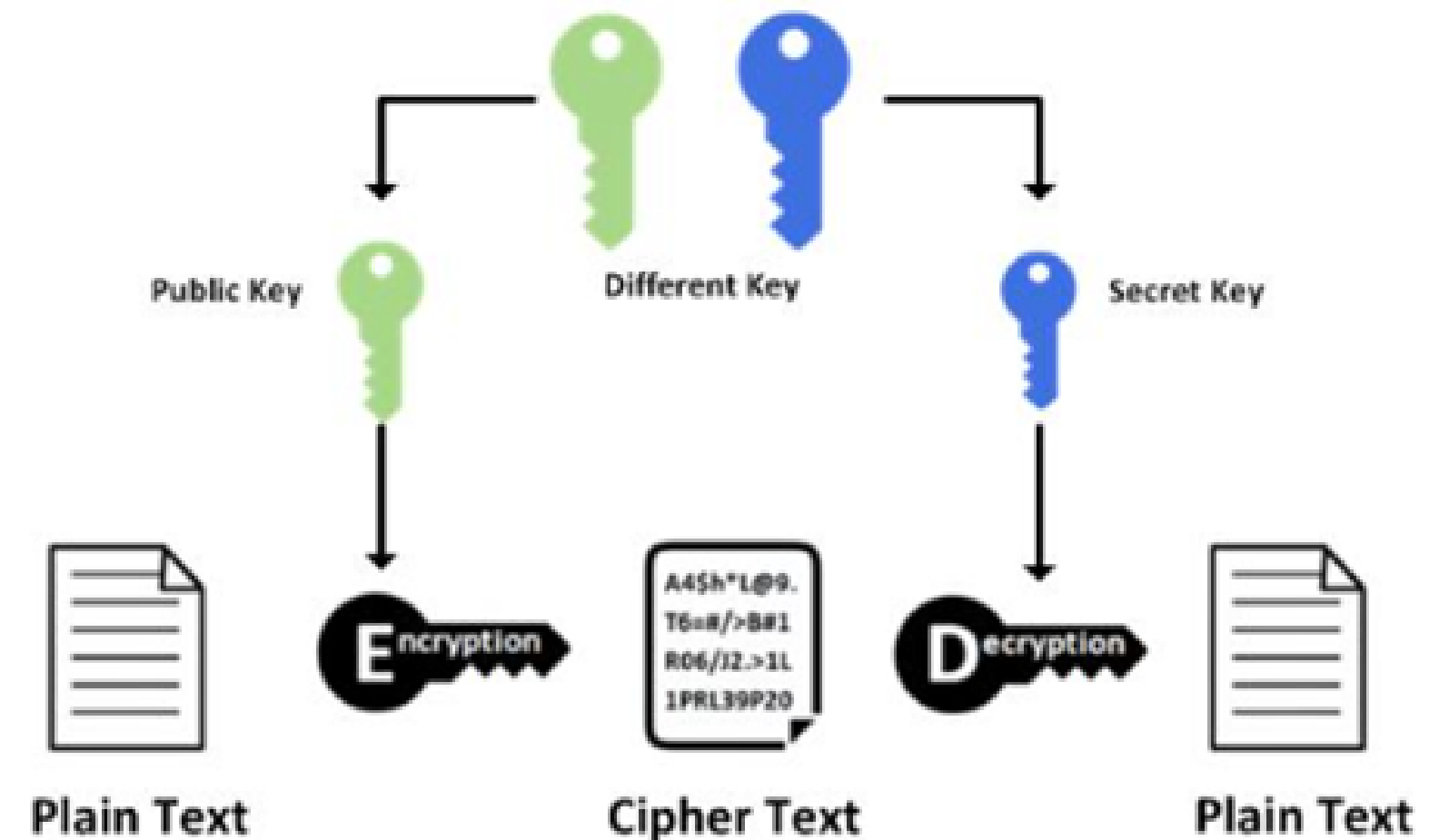
Public-key cryptography คือ การเข้ารหัสแบบกุญแจสาธารณะ หรือบางครั้งเรียกว่า asymmetric cryptography ซึ่งจะตรงข้ามกับ symmetric cryptography

Symmetric vs Asymmetric encryption

Symmetric Encryption



Asymmetric Encryption



Cryptographic techniques used in blockchain (cont.)

2) Hash function

อัลกอริทึมหรือฟังก์ชันที่รับข้อมูลอะไรก็ตามในรูปแบบใดก็ตามและสร้างเอาต์พุต (output) เป็นสตริงความยาวคงที่ โดยฟังก์ชันแฮชจะทำงานโดยรับข้อมูลเข้าและผ่านกระบวนการทางคณิตศาสตร์เพื่อสร้างแฮชของข้อมูลนั้น ๆ

มีคุณลักษณะดังนี้

- ไม่ว่า input จะมีขนาดเท่าไร output จะมีขนาดเท่ากัน
- ไม่สามารถย้อนกลับข้อมูลเดิมจากข้อมูลที่ถูกเข้ารหัส (one-way function)

Hashing Algorithm in Blockchain



<https://medium.com/dcen/hashing-algorithm-in-blockchain-367f09e043cf>



**Continue get
token**

Workshop

Get free token from

1) Sepolia faucet

2) Chainlink faucet

Workshop

!!! Make sure you have 2 Metamask wallets

Connect and get free token from testnet

- 1) Sepolia faucet (SepoliaETH)
- 2) Mumbai faucet (Matic)
- 3) ChainLink (Link and ETH from each testnet)

[Link to ChainLink documentation](#)

Workshop

!!! Make sure you have 2 Metamask wallets

4) OpenZeppelin

5) Remix IDE

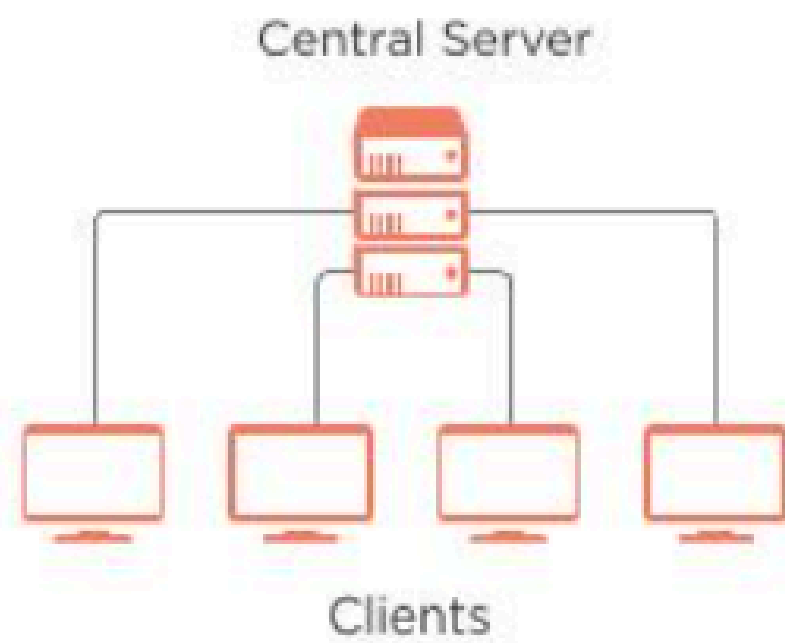
6) Deploy contract to Sepolia Testnet

7) Check block explorer

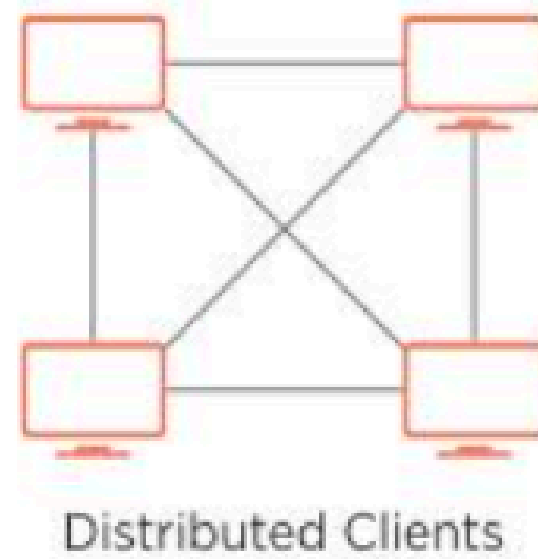


BLOCKCHAIN TECHNOLOGY

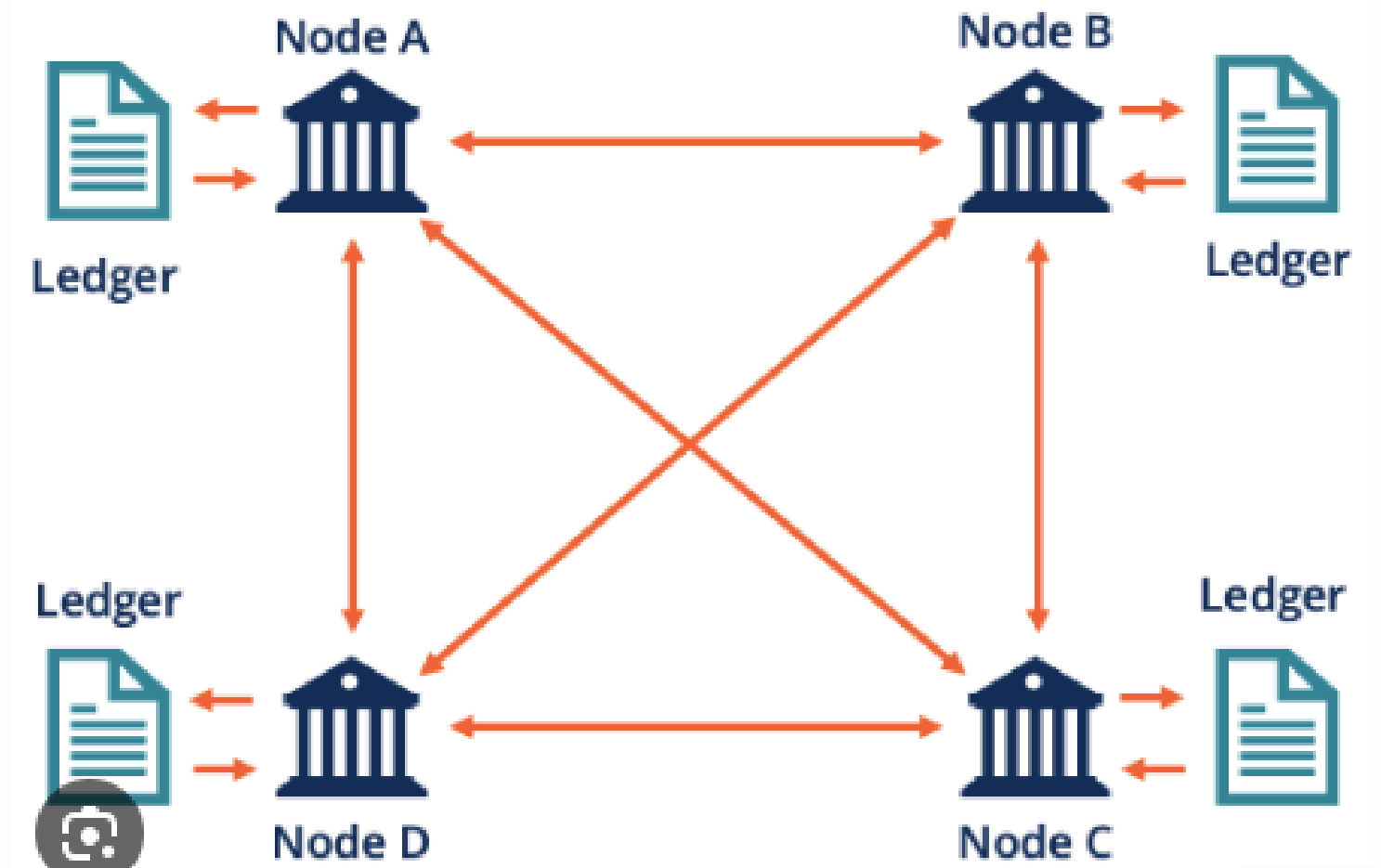
CLIENT/SERVER

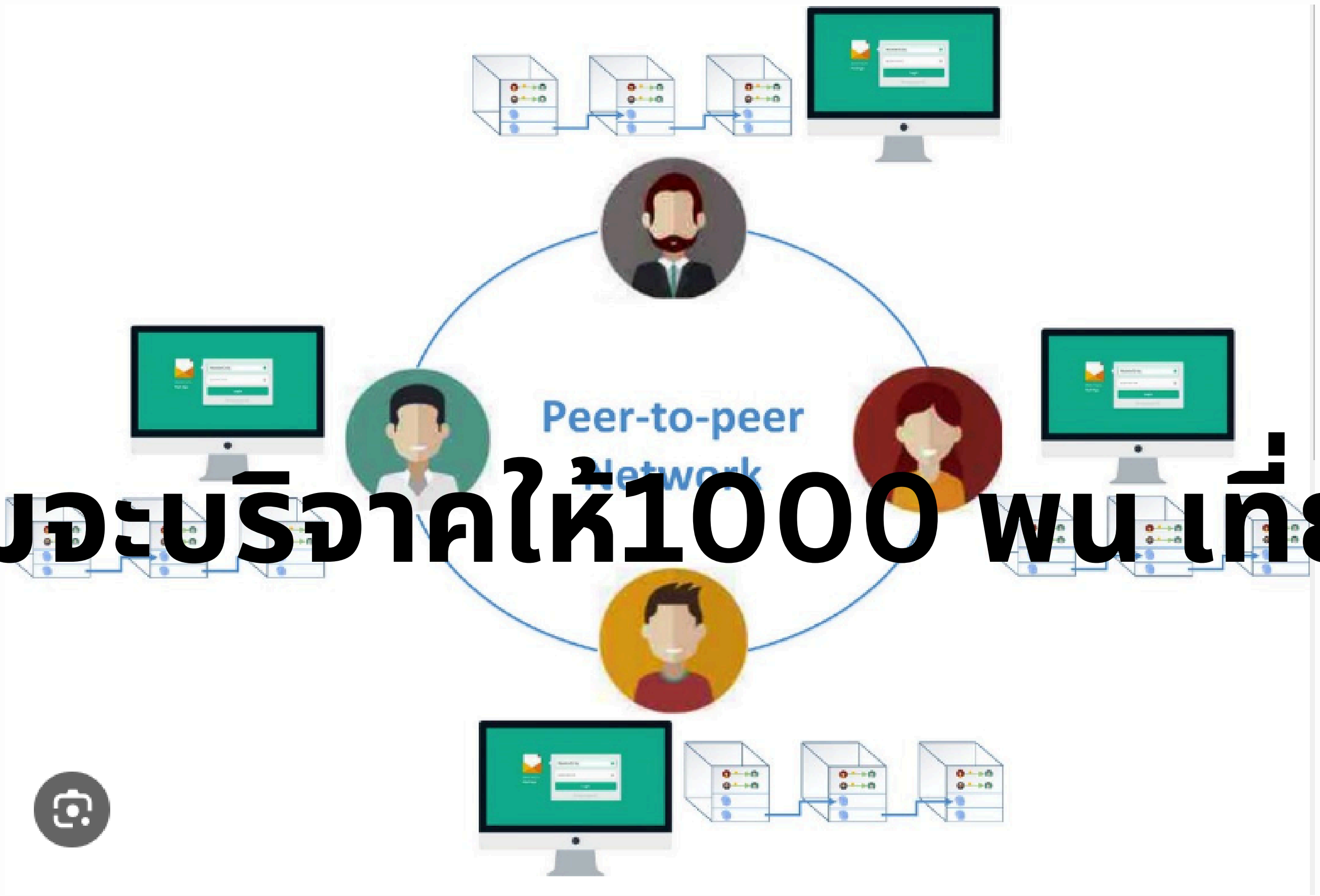


PEER TO PEER



Distributed Ledgers



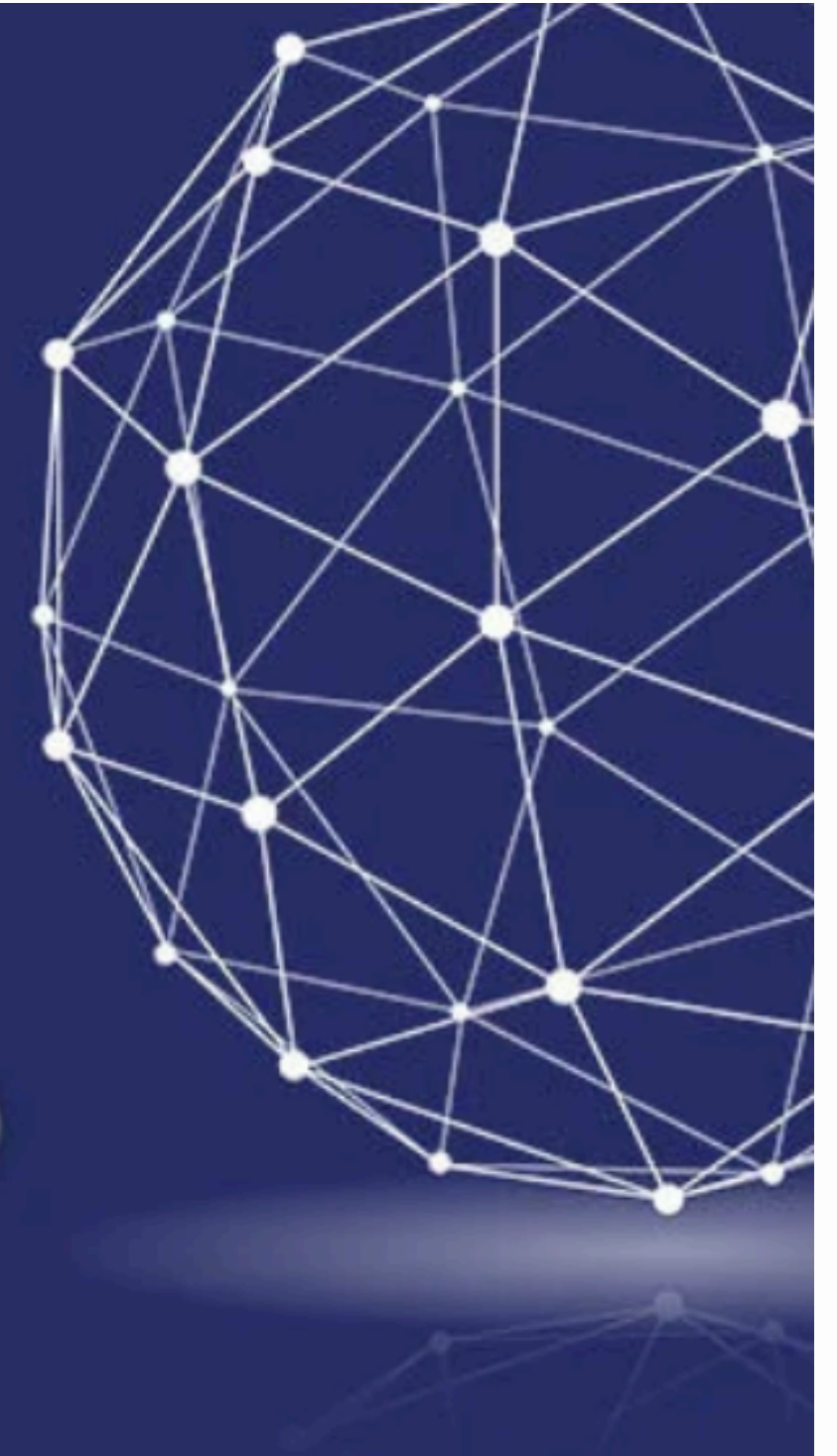
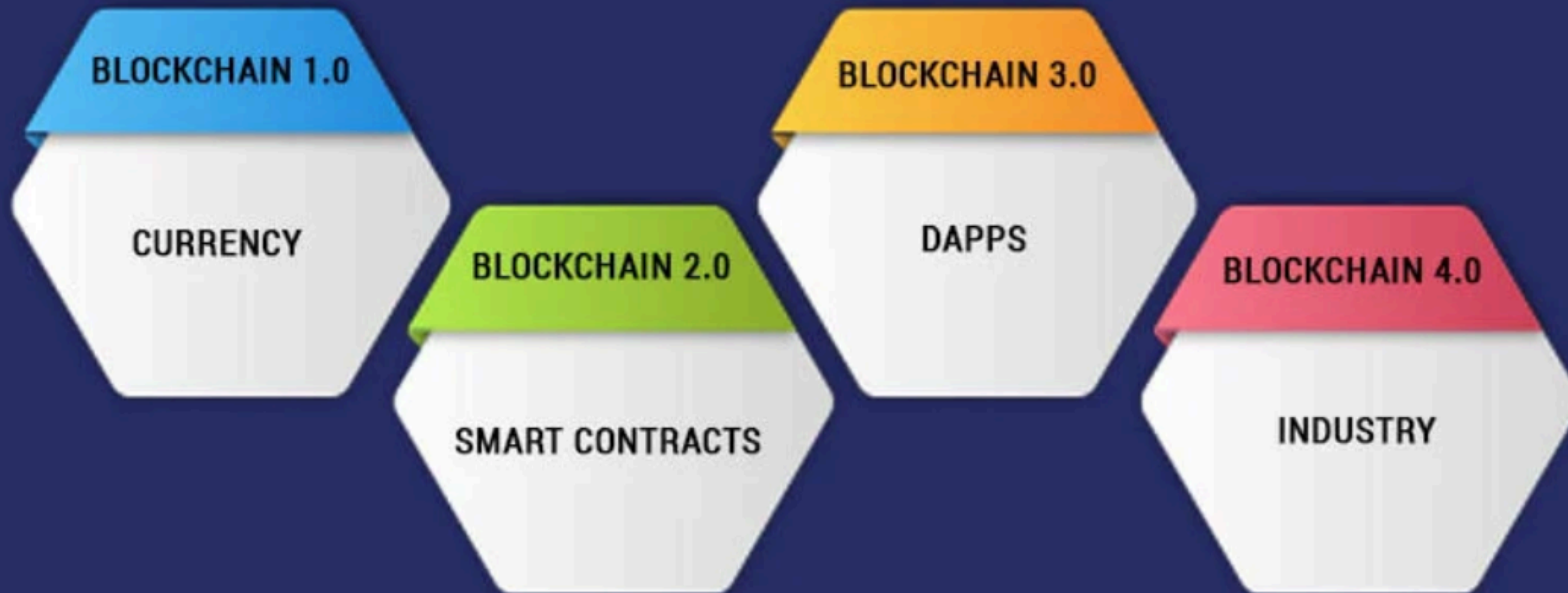
A diagram illustrating a Peer-to-peer Network. In the center, four stylized human avatars (a man with a beard, a man, a woman, and a man) are arranged in a circle, connected by a blue circular line. The text "Peer-to-peer Network" is written in blue across the center. Surrounding this central circle are four computer monitors, each displaying a green screen with a white interface. Each monitor is connected to a series of three server racks, represented by blue and white cubes with colored dots. The entire diagram is set against a light gray background.

ผมจะบริจาคให้ 1000 คน เท่านั้น

Blockchain

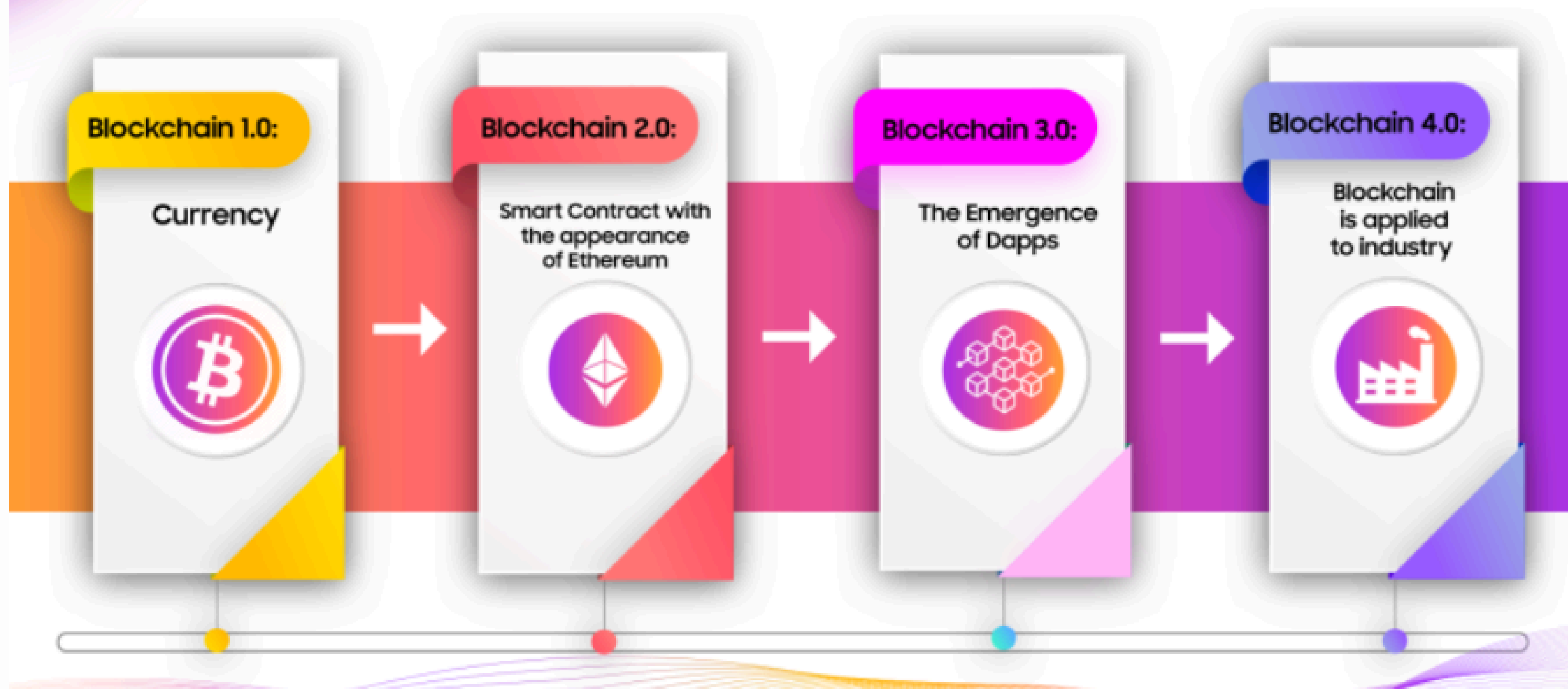
ข้อมูลที่ได้รับการจัดเก็บใน blockchain จะไม่สามารถเปลี่ยนแปลงหรือแก้ไขได้ (immutability) จึงทำให้เทคโนโลยีนี้เป็นที่ยอมรับว่าเป็นเทคโนโลยีสำหรับการจัดเก็บข้อมูลที่มีความปลอดภัย (security) โปร่งใส (transparency) และมีความน่าเชื่อถือสูง (reliability)

BLOCKCHAIN HISTORY AND EVOLUTION



<https://www.ifourtechnolab.com/blog/blockchain-history-and-evolution>

Blockchain Evolution



Search Blockchain, Transactions, Addresses and Blocks

Tether/USD 1.00 ▲0.00%

XRP/USD 0.79 ▼4.35%

Binance Coin/USD 243.93 ▲1.05%

Wise Token/USD 47.45 ▲0.05%



Bitcoin BTC

Bitcoin (BTC) is a decentralized currency that eliminates the need for central authorities such as banks or government to-peer internet network to confirm transactions directly between users.

Price History

\$29,887.19 • 11:10

Vol 26 BTC

1D 1W 1M 1Y MAX USD



Blockchain.com

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DeFi

Academy

News

Developers

Wallet

Exchange

Bitcoin

Ethereum

Bitcoin Cash

Search Blockchain, Transactions, Addresses and Blocks

Q

⚙

Sign

OKB/USD 42.91 ▼0.04%

Dogecoin/USD 0.07 ▲2.43%

Stellar/USD 0.17 ▼3.01%

Chainlink/USD 8.31 ▲19.96%

Aptos/USD 7.49 ▼0.68%

Polygon/USD 0.77 ▲0.76%

TRON/USD 0.12 ▼0.01%

Blockchain

#799610

#799609

#799608

#799607

#799606

#799605

#799604

#799603

#799602

#799601

#799600

#799599

#799598

Market Info

Market Cap ⓘ
\$580,898,858,987

Diluted Market Cap ⓘ
\$627,674,249,309

Vol 24h ⓘ
\$587,736,810

Vol / Market Cap ⓘ
0.0010

24h Change ⓘ
-0.26%

1h Change ⓘ
-0.05%

24h High ⓘ
\$30,419.86

24h Low ⓘ
\$29,569.02

Latest Block
799,606

Block Reward
6.25

Hashes Per Second
-

Circulating Supply ⓘ
19,435,043

Max Supply ⓘ
21,000,000

Consensus ⓘ
Proof of Work

Algorithm ⓘ
SHA-256

Genesis Block Date ⓘ
2009-01-03

Bitcoin Block 0

Mined on January 04, 2009 01:15:05 • [All Blocks](#)

- Satoshi
- Notable Block

Coinbase Message •  EThe Times 03/Jan/2009 Chancellor on brink of second bailout for banks

Bitcoin Genesis

On January 3rd 2009, the Bitcoin network was created when Satoshi Nakamoto (the project's mysterious creator) mined the “Genesis” block. The 50 bitcoin coinbase reward is unredeemable, as it was omitted from the transaction database. This means any attempt to spend it would be rejected by the network. Whether this was intentional or not still remains unknown.

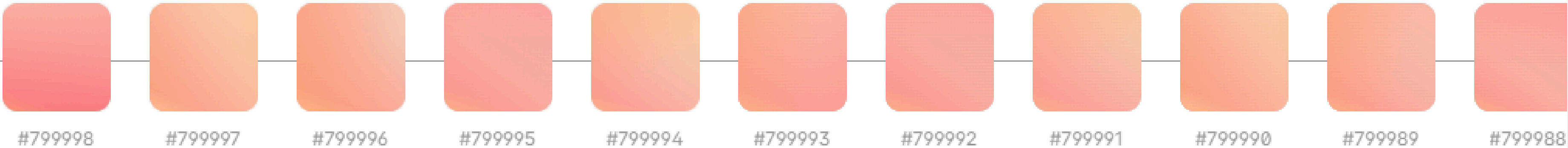
A total of 0.00 BTC (\$0.00) were sent in the block with the average transaction being 0.0000 BTC (\$0.00). Satoshi earned a total reward of 50.00 BTC \$0.00. The reward consisted of a base reward of 50.00 BTC \$0.00 with an additional 0.0000 BTC (\$0.00) reward paid as fees of the 1 transactions which were included in the block.

Details

Hash	00000-ce26f 	Depth	799,998
Capacity	0.03%	Size	285
Distance	14y 6m 19d 8h 49m 45s	Version	0×1
BTC	0.0000	Merkle Root	4a-3b 
Value	\$0.00	Difficulty	1.00
Value Today	\$0.00	Nonce	2,083,236,893
Average Value	0.00000000000 BTC	Bits	486,604,799
Median Value	50.000000000 BTC	Weight	1,140 WU
Input Value	0.00 BTC	Minted	50.00 BTC
Output Value	50.00 BTC	Reward	50.000000000 BTC
Transactions	1	Mined on	Jan 04, 2009, 1:15:05 AM
Witness Tx's	0	Height	0
Inputs	1	Confirmations	799,998
Outputs	1	Fee Range	0-0 sat/vByte
Fees	0.000000000 BTC	Average Fee	0.000000000
Fees Kb	0.00000000 BTC	Median Fee	0.000000000
Fees kWU	0.00000000 BTC	Miner	Satoshi

Blockchain

→



Transactions





0 ID: **4a5e-a33b** 

1/04/2009, 01:15:05

From Block Reward

To **1A1z-vfNa** 

50.00000000 BTC • \$1,488,830

Fee 0 Sats • \$0.00

^

From

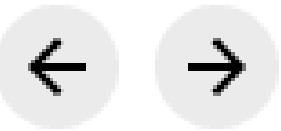
 1

Block Reward
0.00 BTC • \$0.00

To

1 **1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa**  

50.00000000 BTC • \$1,488,830



Bitcoin Block 799,998

Mined on July 24, 2023 10:03:53 • [All Blocks](#)

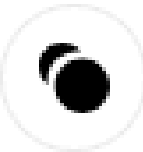
M4

Coinbase Message •     Q:  t     

A total of 13,914.86 BTC (\$414,652,759) were sent in the block with the average transaction being 4.4800 BTC (\$133,500). M4 earned a total reward of 6.25 BTC \$186,245. The reward consisted of a base reward of 6.25 BTC \$186,245 with an additional 0.2036 BTC (\$6,067.13) reward paid as fees of the 3,106 transactions which were included in the block.

Details

Hash	00000-2b5b4 	Depth	1
Capacity	158.54%	Size	1,662,401
Distance	14y 6m 19d 8h 54m 2s	Version	0×310d0000
BTC	13,914.8630	Merkle Root	39-96 
Value	\$414,652,759	Difficulty	53,911,173,001,054.59
Value Today	\$414,172,000	Nonce	1,394,660,396
Average Value	4.4799945213 BTC	Bits	386,218,132
Median Value	0.00165310 BTC	Weight	3,992,687 WU
Input Value	13,915.07 BTC	Minted	6.25 BTC
Output Value	13,921.32 BTC	Reward	6.45357181 BTC
Transactions	3,106	Mined on	Jul 24, 2023, 10:03:53 AM
Witness Tx's	2,826	Height	799,998
Inputs	6,911	Confirmations	1
Outputs	8,642	Fee Range	0-461 sat/vByte
Fees	0.20357181 BTC	Average Fee	0.00006554
Fees Kb	0.0001225 BTC	Median Fee	0.00002047
Fees kWU	0.0000510 BTC	Miner	M4

	0 ID: f31c-bb92  7/24/2023, 10:03:53	From Block Reward To 2 Outputs	6.45357181 BTC • \$192,311 Fee 0 Sats • \$0.00	▼
TX	1 ID: d4d3-c640  7/24/2023, 09:52:07	From bc1p-ytjr  To 2 Outputs	0.00151686 BTC • \$45.20 Fee 94.5K Sats • \$28.17	▼
TX	2 ID: 10b3-90fc  7/24/2023, 09:52:11	From bc1p-uq8m  To 2 Outputs	0.00015208 BTC • \$4.53 Fee 105.8K Sats • \$31.53	▼
TX	3 ID: a1c9-1a3b  7/24/2023, 09:51:18	From bc1q-vzej  To 2 Outputs	0.13624316 BTC • \$4,059.95 Fee 90.0K Sats • \$26.82	▼
TX	4 ID: 796b-f69d  7/24/2023, 09:47:33	From bc1q-vzej  To 2 Outputs	0.06896315 BTC • \$2,055.05 Fee 60.0K Sats • \$17.88	▼
TX	5 ID: ede1-3fd7  7/24/2023, 09:54:50	From bc1q-pnuy  To bc1q-6yej 	0.06845185 BTC • \$2,039.82 Fee 33.0K Sats • \$9.83	▼
TX	6 ID: d61c-35dc  7/24/2023, 09:56:58	From bc1q-j0h2  To 2 Outputs	1.99957700 BTC • \$59,585.93 Fee 42.3K Sats • \$12.61	▼
TX	7 ID: f854-586e  7/24/2023, 09:54:19	From bc1q-mnjq  To 2 Outputs	552.24476474 BTC • \$16,456,490 Fee 40.3K Sats • \$12.01	▼

Transactions

Last

First

Value

Value

Fee

Fee

0 ID: f31c-bb92

7/24/2023, 10:03:53

From Block Reward

To 2 Outputs

6.45357181 BTC • \$192,311

Fee 0 Sats • \$0.00

From	To
1 Block Reward 0.00 BTC • \$0.00	1 3KZDwmJHB6QJ13QPXHaW7SS3yTESFPZoxb 6.45357181 BTC • \$192,311 2 Unknown 0.00000000 BTC • \$0.00

TX

1 ID: d4d3-c640

7/24/2023, 09:52:07

From bc1p-ytjr

To 2 Outputs

0.00151686 BTC • \$45.20

Fee 94.5K Sats • \$28.17

From	To
1 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2... 0.00246230 BTC • \$73.37	1 bc1pv6ar4n5ex7g0sf8h667xmlv0g6k8d5j2zrlwgz343scnxkxpl0... 0.00121000 BTC • \$36.06 2 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2sh... 0.00030686 BTC • \$9.14

TX

2 ID: 10b3-90fc

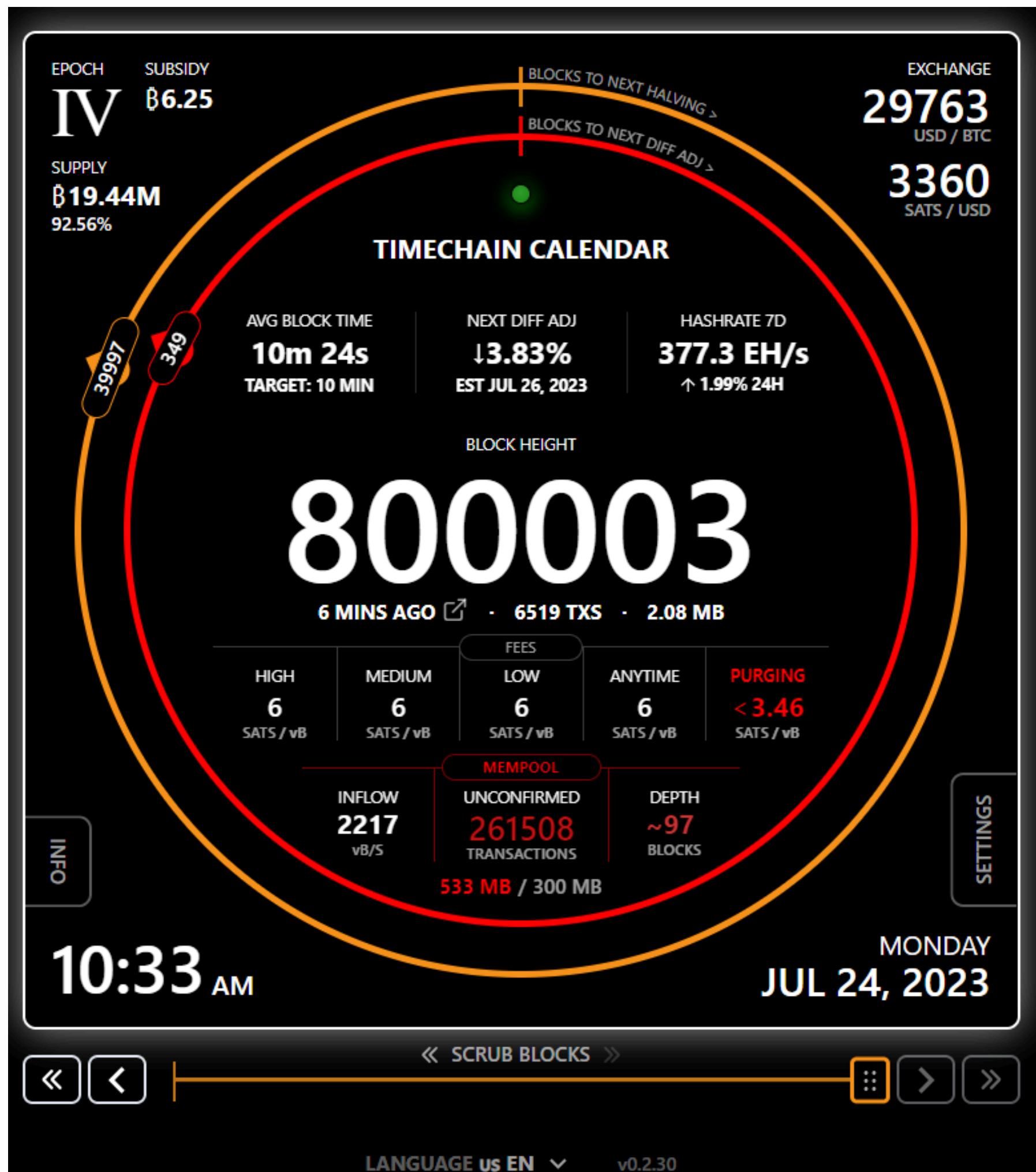
7/24/2023, 09:52:11

From bc1p-uq8m

To 2 Outputs

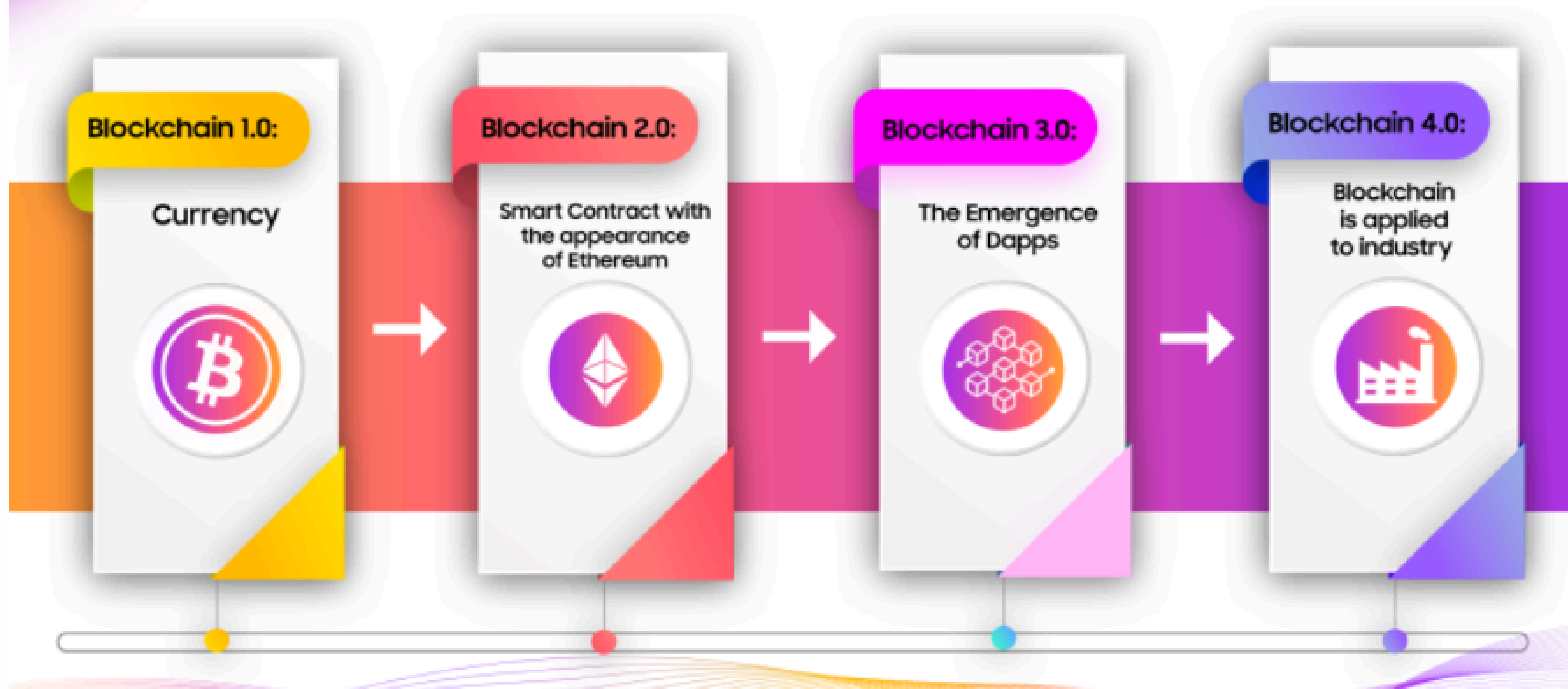
0.00015208 BTC • \$4.53

Fee 105.8K Sats • \$31.53



[Link to Timechain Calendar](#)

Blockchain Evolution



Blockchain 1.0

Blockchain 1.0:

Currency



Transactions

⬆️ Last **First** ⬆️ Value ⬆️ Value ⬆️ Fee ⬆️ Fee

0 ID: f31c-bb92 7/24/2023, 10:03:53		From Block Reward To 2 Outputs		6.45357181 BTC • \$192,311 Fee 0 Sats • \$0.00
From		To		
← 1 Block Reward 0.00 BTC • \$0.00		1 3KZDwmJHB6QJ13QPXHaW7SS3yTESFPZoxb 6.45357181 BTC • \$192,311 2 Unknown 0.00000000 BTC • \$0.00		
TX 1 ID: d4d3-c640 7/24/2023, 09:52:07		From bc1p-ytjr		0.00151686 BTC • \$45.20 Fee 94.5K Sats • \$28.17
From		To		
← 1 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2... 0.00246230 BTC • \$73.37		1 bc1pv6ar4n5ex7g0sf8h667xmlv0g6k8d5j2zrlwgz343scnxkxpl0... → 0.00121000 BTC • \$36.06 2 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2sh... 0.00030686 BTC • \$9.14		
TX 2 ID: 10b3-90fc 7/24/2023, 09:52:11		From bc1p-uq8m		0.00015208 BTC • \$4.53 Fee 105.8K Sats • \$31.53

ແນະນຳ Lightning Wallet ມາໃຊ້ງານ

https://www.walletofsatoshi.com

WALLET OF SATOSHI

Screenshots FAQ Contact Media Download App

9:26

WALLET OF SATOSHI

35,529 sats
about \$13

Have a 🍷 on me! 🥰
8 hours ago +15,000 sats
about \$5

History

Receive Send

The world's simplest Bitcoin Lightning Wallet

Download on the App Store GET IT ON Google Play

Number of lightning payments made:

10,381,069

Lightning Address ✕

WALLET OF SATOSHI

Copy Share

glibdoll72@walletofsatoshi.com

Customize

@ ₿ ⚡

Workshop 1

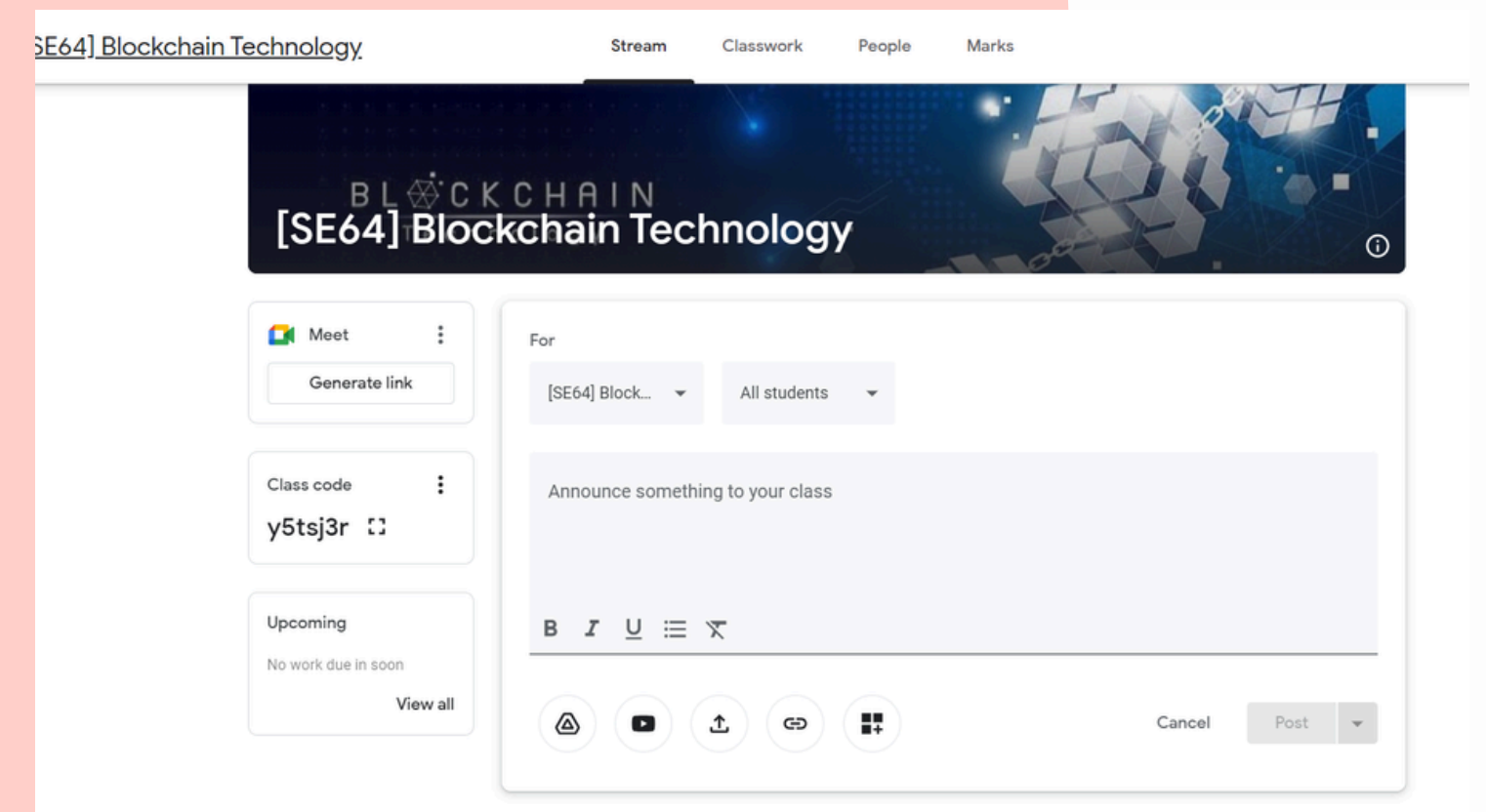
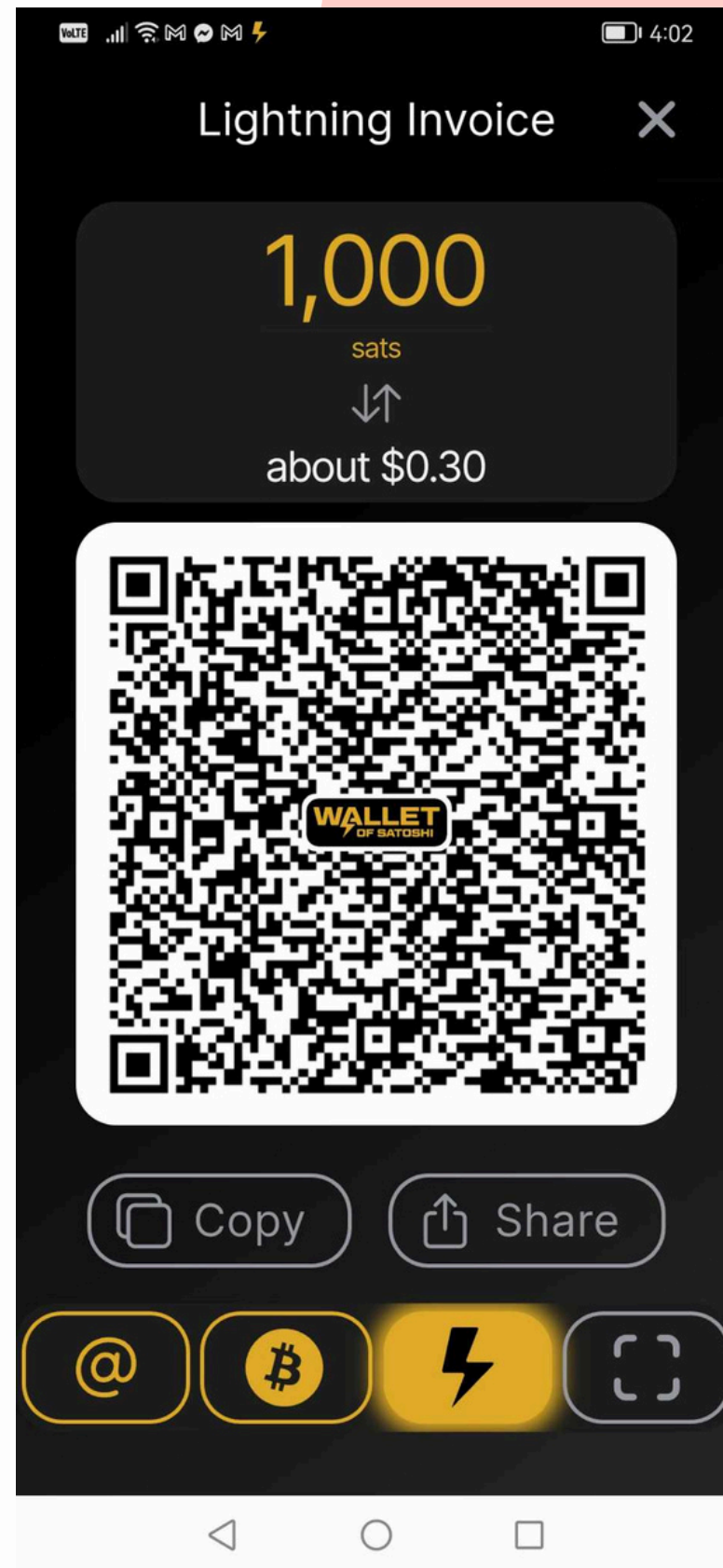
Bitcoin lightning network

- 1) Download App "Wallet of Satoshi"
- 2) Register in application
- 3) Generate lightning invoice 100 sats (100 satoshi)
- 4) Wait till get a lightning invoice QR code

Workshop 1

Bitcoin lightning network

Upload to
GClassroom





Bitcoin Moves Money Forward | Bitcoin Halving

Coinbase

Bitcoin move ney forward



Watch on  YouTube 



What is a Bitcoin Halving? Explained by CoinGecko

CoinGecko

Bitcoin Halving



Watch on  YouTube



Bitcoin Halving Explained (Animated) - Why, How, and What it Means for You!

Whiteboard Crypto



Watch on  YouTube